

# Mehmet Aygun

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## EDUCATION

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- **Technical University of Munich (TUM)** Munich, Germany  
*Master of Science in Computer Science* Oct. 2018 – Dec. 2020
- **Istanbul Technical University (ITU)** Istanbul, Turkey  
*Bachelor of Science in Computer Science* Sep. 2013 – June. 2017

## RESEARCH EXPERIENCE

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- **TUM Computer Vision Lab** Munich, DE  
*Research Student* Oct 2018 - Current
  - **Holistic 4D Scene Analysis** : Working on 4D scene analysis using Lidar data. Particularly on semantic/instance segmentation and multi object tracking for autonomous driving scenario. Advisors: Prof. Laura Leal-Taixe, Dr. Aljosa Osep.
  - **3D Shape Analysis**: Worked on shape correspondence problem with self-supervised deep learning methods using functional maps and heat kernels. Advisors: Zorah Lahner, Prof. Daniel Cremers.
- **ITU Computer Vision Lab** Istanbul, Turkey  
*Research Student* Dec 2013 - June 2017
  - **Medical Image Segmentation**: Worked on Brain Tumor segmentation using convolutional neural networks under supervision of Prof. Gozde Unal. Conducted research for designing CNN architectures for multi modal 3D MRI images.
  - **Transfer Learning**: Worked on transfer learning for convolutional neural networks. Collaborated with Dr. Yusuf Aytar from MIT CSAIL and proposed a statistical regularization method for transferring knowledge between different CNN architectures.
  - **Facial Image Analysis**: Worked on facial image analysis using both traditional approaches and deep learning under supervision of Assoc. Prof. Hazim Ekenel. Developed real time age and gender recognition systems. Proposed a method for apparent age estimation.

## PUBLICATIONS

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- **Conference Papers:**
  - **M. Aygun**, A. Osep, M.Weber, M.Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, "4D Panoptic Lidar Segmentation", In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021
  - **M. Aygun**, Z. Lahner, D. Cremers, "Unsupervised Dense Shape Correspondence using Heat Kernels", In International Conference on 3D Vision (3DV), 2020
- **Workshop Papers & Preprints:**
  - **M. Aygun**, Y. Aytar, H. K. Ekenel, "Exploiting Convolution Filter Patterns for Transfer Learning", ICCV TASK-CV workshop, 2017
  - R. C. Malli\*, **M. Aygun\***, H. K. Ekenel, "Apparent Age Estimation Using Ensemble of Deep Learning Models", CVPRW, June 2016 (\*:equal contribution.)
  - **M. Aygun**, Y. Sahin, G. Unal, "Multi modal Convolutional Neural Networks for Brain Tumor Segmentation, arXiv preprint arXiv:1809.06191 (2018)

## HONORS & AWARDS

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- **ICCV 17 Workshop, Honourable Mention Award**: Received Honourable Mention Award at ICCV 17 TASK-CV workshop for our paper 'Exploiting Convolution Filter Patterns for Transfer Learning'.
- **DAAD scholarship**: 2 year master scholarship for studying in Germany.
- **Bachelor Thesis Award**: Best Bachelor Thesis Award from Computer Engineering Department for my thesis titled "Multi Modal 3D Convolutional Neural Networks for Brain Tumor Segmentation"
- **Undergraduate Research Scholarship** : Research scholarship from science and technology research council of Turkey during my last two year of bachelor education.

## PROFESSIONAL EXPERIENCE

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- **Coriolis Pharma** Munich, Germany  
*Computer Vision Engineer* *Sep 2019 - March 2020*
  - **Biological image analysis:** Developed methods for particle analysis using deep learning based classification, instance segmentation and unsupervised learning methods. Used frameworks: Tensorflow, Keras, OpenCV.
- **Motius** Munich, Germany  
*Machine Learning Engineer* *Jan 2019 - April 2019*
  - **Handwritten Text Recognition:** Developed handwritten text recognition system using deep learning (CNN+LSTM+CTC). Reduced word error rate from 18% to 12%. Used frameworks: Tensorflow, OpenCV.
- **Bunsar** Istanbul, Turkey  
*Computer Vision Engineer* *Feb 2017 - Sep 2018*
  - **Visual Search Engine from Images and Videos:** Research and development activities for image retrieval system using deep learning methods. Topic studied: Object Detection, Image Segmentation, Image Classification, Large Scale Image Retrieval Used frameworks: Caffe, OpenCV, Sklearn

## SKILLS

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- **Programming:** Python, C/C++, Matlab
- **Frameworks and Tools:** Pytorch, Tensorflow, Caffe, OpenCV, Sklearn, Pandas, SQL, Numpy, Vim, Linux, Docker, Kubernetes

## RELATED COURSEWORK

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- **TUM:** Natural Language Processing, Deep Learning, Variational Methods, Multiple View Geometry, Tracking and Detection
- **ITU:** Computer Vision, Learning from Data, Probability, Linear Algebra, Calculus I & II

## OTHER

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- **Reviewer:** Reviewed 8 papers for MICCAI Deep Learning Workshop, 2017-2018
- **Volunteer:** Volunteering work for high school student at code.org

## REFERENCES

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- **Prof. Laura Leal-Taixe:** leal.taixe@tum.de
- **Prof. Gozde Unal:** gozde.unal@itu.edu.tr
- **Prof. Hazim Kemal Ekenel:** ekenel@itu.edu.tr